

Steve Gwade

Machine Learning R&D Engineer

Yaounde, Cameroon · Open to Relocation

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Professional Summary

Machine Learning R&D Engineer with 2.5+ years of cumulated experience developing machine learning systems across computer vision, generative AI, healthcare, agriculture, finance, and assistive technologies. Core contributor to the LINA Assistive Vision System, awarded 1st Place Worldwide in the Huawei ICT Competition 2025 Innovation Track. Experienced in deep learning, multi-task learning, retrieval-augmented generation (RAG), and real-time inference systems, with hands-on expertise spanning data pipelines, model training, evaluation, deployment, and backend integration. Passionate about transforming research ideas into reliable products that solve practical problems at scale.

TECHNICAL SKILLS

- **Machine Learning Research:** Supervised Representation Learning, Multi-Task Optimization, Contrastive Learning, Cross-Validation, Statistical Error Bounds, Hyperparameter Tuning, Multi-Modal Systems, Feature Engineering.
- **Generative AI & NLP:** Large Language Models (LLMs), Advanced RAG (Hybrid Dense/Sparse Retrieval, Vector Search, FAISS), Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, Structured Outputs (Pydantic/JSON Schema), Online A/B Testing.
- **Computer Vision:** Deep Object Detection, mAP Benchmarking, Image Classification Backbones, Semantic Segmentation, Real-Time Edge Quantization (INT8 Calibration, TensorRT Optimization), CNNs.
- **MLOps & Infrastructure:** MLflow Run Tracking, Model Monitoring, Containerized Environments (Docker, Docker Compose), AWS Architecture (EventBridge, SQS), GitHub Actions CI/CD Frameworks, Data Pipeline Engineering.
- **Frameworks & Libraries:** PyTorch, Scikit-Learn, NumPy, Pandas, Hugging Face Transformers, LangChain, Core SQL, Redis Distributed Caching.
- **Methodologies & Domains:** Agile/Scrum Development, Technical Stakeholder Alignment, Algorithmic Analysis, Competitive Programming.

WORK EXPERIENCE

AI Systems Engineer

MITS SARM – Moabi B2B Travel Agency

Apr 2026 – Present

Yaounde, Cameroon

- Architected and deployed an asynchronous, LLM-driven engine utilizing instructor-guided structured outputs (JSON Schema validation) via Meta API webhooks, reducing conversational stream processing latency by 90% ($P_{99} < 120$ ms).
- Designed high-concurrency, multi-turn dialog context layers using Redis distributed caching and DynamoDB to securely track long-term history states across 500+ concurrent user streams.
- Integrated generative workflows into legacy systems by engineering decoupled, event-driven asynchronous inference pipelines using AWS EventBridge and SQS queues, managing live traffic spikes up to 4x baseline capacity.
- Led an AI engineering pod of 2 developers to build automated GitHub Actions pipelines for automated LLM checkpoint evaluation and safety validation, enforcing a 95%+ evaluation threshold while cutting delivery cycles by 40%.

Research Engineer (AI/ML) (Part-time)

JIANTS – Systems Intelligence Lab

Jan 2025 – Jan 2026

Douala, Cameroon

- Profile-tuned 4 multi-modal foundational architectures (Vision, OCR, STT, TTS), reducing prototype-to-production deployment latency windows by 30% across target hardware environments.
- Developed reproducible PyTorch data pipelines integrating stochastic geometric and photometric transforms with automated model training telemetry via systematic MLflow run-tracking.
- Quantized LINA spatial perception layers using TensorRT INT8 post-training protocols, boosting execution edge throughput by 56% while preserving baseline model validation accuracy bounds.
- Conducted compute-efficiency profiling (FLOPs, VRAM footprint, memory bandwidth) for 3 multi-modal variants, presenting empirical benchmarks to technical stakeholders to guide hardware execution roadmaps.

Machine Learning & Software Engineer (Part-time)

TGNix – IT Consulting

Mar 2024 – Dec 2024

Douala, Cameroon

- Productionalized predictive decision-support scoring models within 2 enterprise ERP modules, introducing automated model monitoring to reduce manual operational review overhead metrics by 30%.
- Accelerated automated decision-engine execution times by 25% via targeted SQL query optimization, localized data caching layers, and database performance stress-testing.
- Built high-throughput FastAPI REST endpoints for distributed model inference, isolating asynchronous feature workloads across 3 decoupled microservices following domain-driven design boundaries.
- Containerized multi-tiered application layers using Docker and Docker Compose, achieving isolated runtimes that eliminated environmental execution drift across staging and production environments.

AI Systems Engineer (Internship)

Jun 2023 – Nov 2023

- Developed Python APIs to serve 5 core internal predictive ML model components under high-concurrency conditions, managing versioning and system interface contracts.
- Built modular streaming data pipelines with LangChain to automate structural text parsing and semantic answering across centralized documentation datastores.
- Fine-tuned a YOLOv8 object detection architecture across 4 crop health datasets, optimizing anchor box allocations to secure sub-5 ms edge inference speeds and improve mAP by 4.2%.
- Executed automated feature engineering and data curation cycles, auditing annotation metrics for 10k+ images while applying rigid validation splitting rules to eliminate evaluation dataset leakage.

RESEARCH & SELECTED PROJECTS

LINA – Assistive Vision System for the Blind

Huawei ICT Competition 2025

- Co-developed an international multi-modal assistive ecosystem combining real-time object detection, OCR, and spatial scene comprehension into a low-latency execution pipeline.
- Mitigated presentation-attack vulnerabilities by training binary face anti-spoofing classifiers, implementing rigorous security testing to intercept spoof vectors across variable environmental testing conditions.
- **Awarded 1st Place Worldwide (Innovative Track)** among international entrants for exceptional model compression, serving efficiency, and collaborative architectural novelty.

Multi-Task Transfer Learning for Malaria Diagnosis

MSc Research Thesis, 2025

- Formulated a joint multi-task deep learning problem in PyTorch to execute simultaneous semantic cell segmentation and parasite detection from thin blood smears.
- Evaluated negative transfer optimization dynamics across distinct model heads, establishing a stable joint topology yielding a 97% macro F_1 -score.
- Published a reproducible open-source research repository featuring pipeline configuration matrices, execution scripts, and verified pre-trained model weights.

VERO – Research Workspace for Grounded AI

Open-Source Project, 2025

- Engineered an advanced RAG environment featuring deterministic text parsing, semantic chunking, and cross-reference metadata generation across document source trees.
- Integrated dense vector retrieval with BM25 keyword matching, optimized via Reciprocal Rank Fusion (RRF) and Cross-Encoder reranking models to maximize retrieval relevance (NDCG@K).

PlumVision – Real-Time Plum Quality Control Engine

JCIA Hackathon Winner 2025

- Developed an ultra-lightweight (~ 3 MB) ConvNeXt image classifier using representation learning, registering a 98% accuracy profile on target evaluation datasets.
- Exposed model parameters via optimized FastAPI services, maintaining operational inference latency under 12 ms to support continuous hardware assembly integrations through stress-testing procedures.

EDUCATION

MSc in Engineering (Data Science & AI)

2023 – 2025

National Higher Polytechnic School of Douala

Douala, Cameroon

- Graduation Status: Graduated with Distinction (Top 5% of cohort, **GPA: 3.6/4**).
- Core: Mathematical Optimization, Probabilistic Machine Learning, Distributed Deep Learning, Statistical Evaluation.
- Thesis Repository Link: github.com/GwadeSteve/MTTL-Research-Malaria

BSc in Engineering (Computer Science)

2020 – 2023

National Higher Polytechnic School of Douala

Douala, Cameroon

- Core Modules: Problem Solving, Algorithmic Complexity, Advanced Data Structures, Linear Algebra, Mathematical Statistics, Systems Engineering.

AWARDS & RECOGNITION

- **1st Place Worldwide Winner**, Huawei ICT Competition 2025 (Innovative Track) – Core Developer for the LINA Assistive Vision project.
- **Ranked 2nd Across Africa**, Orange Digital Center Champions League 2025 – Algorithmic and Competitive Programming tournament (CodinGame).
- **Winner, Best AI Solution** for Optimization and Efficiency, JCIA 2025 Track – Awarded for PlumVision real-time throughput and deployment speed.
- **Winner, Most Innovative AI Prototype**, JCIA 2025 Track – Awarded for EatWise, a multi-modal computer vision framework for targeted nutritional parsing.